

CHRISTOPHER OOI EN SHEN

Waterloo, Ontario

+1437-665-5281

Email

My Portfolio

LinkedIn

Github

Education

Wilfrid Laurier University

Bachelor of Science in Computer Science

- CGPA of **3.85**

Sep. 2023 – Oct. 2026

Waterloo, Ontario

Work Experience

Hecaton

Contract Software Engineer

Oct. 2025 – Present

Toronto, Ontario

- Building core features and infra for **1Bitcoin.ca**, a Bitcoin exchange processing **\$200M+** across **96,000+** **transactions**: instant BTC Lightning payouts cutting settlement from **~60 min to seconds**, and AI-driven compliance on a TypeScript/CockroachDB stack.
- Delivered a **production frontend foundation** for **Fleetcraft** (Series A) by owning the **end-to-end UI architecture** and implementing a **hybrid search capability** using BM25 + vector embeddings.
- Shipped a production **Bitcoin notarization platform** processing **500+** monthly transactions, built on a secure **full-stack architecture** with Next.js, Supabase, Prisma, and bitcoinjs-lib.
- Built **reAIGents**, an **AI real estate platform** handling **6M+** property records, saving **\$10–15K** in API costs through cached, rate-limited geocoding and a Next.js/PostGIS architecture supporting spatial queries and real-time notifications.

Manulife

Software Engineer Intern

May. 2026 – Sep. 2026

Waterloo, Ontario

- Built an **AI diagram generation pipeline** using **Django, Celery, Redis, and Azure AI Foundry**, orchestrating **4 model deployments** in a self-correcting agent loop with geometry validation to turn natural language queries into publishable **enterprise architecture diagrams**.
- Deployed the platform to **Azure AKS** through a **GitOps deployment workflow** using **Istio, ExternalSecrets, Workload Identity, and Entra authentication** for Postgres and Azure AI services.

Bolttech

Data Engineer Intern

May. 2025 – Sep. 2025

Kuala Lumpur, Malaysia

- Generated **\$15K+** in **annual savings** by **re-architecting legacy automation** into a **serverless AWS ETL pipeline (Lambda/Step Functions)** scaling to process **1M+** daily inventory records.
- Reduced data processing time by **95% (45m → 2m)** by engineering memory-efficient **Pandas + Databricks** pipelines for large datasets and optimizing transformation patterns to avoid memory bottlenecks.

Sensoft Technologies

Software Engineer Intern

May. 2024 – Sep. 2024

Penang, Malaysia

- Reduced energy costs 15%** by building a hybrid forecasting and reporting pipeline using **Facebook Prophet** for **time-series prediction** and the **OpenAI API** to auto-generate actionable reports.
- Delivered a **real-time React dashboard** monitoring **15,000+ Modbus sensors** at **60 FPS** by implementing **list virtualization** and optimizing **high-frequency WebSocket UI** updates.

Personal Projects

MentorTrader | Next.js, TypeScript, PostgreSQL, Stripe, Supabase, Docker

- Built a two-sided marketplace connecting retail investors with vetted mentors, reaching **120+ mentors and 20K+ monthly visitors**.
- Engineered an **AI mentor-matching wizard** using **OpenAI embeddings, PostgreSQL vector search, and Claude reranking**, with budget-aware recommendations and direct handoff into mentor messaging.

Digital Wealth Management | Next.js, TypeScript, tRPC, Prisma, PostgreSQL

- Built an **Ontario Securities Commission (OSC) approved** Bitcoin custody and BTC-backed lending platform with secure onboarding, deposits/withdrawals, lending, and compliance workflows.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Java, C, Ruby

Frameworks: Next.js, React/Angular, Express, FastAPI, Flask, Rails, Django

Data/Infra: AWS (Lambda, Step Functions, RDS, SQS, S3), Azure, AKS, AKV, Kubernetes, Firebase, Databricks, Docker, Redis, PostgreSQL

Developer Tools: GitHub Actions, Sentry, Vercel, GCP Functions